

Explainable Stock Recommendation System with Volatility Analysis, Forecasting, and Confidence-Interval-Based Suggestions

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Abstract—Investors face an overwhelming number of stocks, making it challenging to choose the right ones for their risk appetite and investment horizon. Traditional recommendation systems often lack transparency, creating a trust gap. This project proposes an explainable stock recommendation system that integrates content-based and collaborative filtering with volatility analysis, forecasting, and confidence-interval-based insights. The system precomputes analytics for real-time response and provides transparent explanations for each recommendation. Such a solution is highly relevant to fintech platforms prioritizing both performance and explainability.

Index Terms—Stock Recommendation, Explainable AI, Forecasting, Volatility, Hybrid Models, Fintech

I. INTRODUCTION AND MOTIVATION

With thousands of publicly traded stocks, investors often struggle to identify options that align with their risk profile and investment horizon. Current recommendation engines are often opaque, offering predictions without justifying the reasoning. This undermines trust in automated financial decision-making systems.

The proposed project addresses this gap by developing a personalized, explainable stock recommendation engine. It provides recommendations based on user risk profiles, sector preferences, and investment horizons, supported by volatility analysis, confidence-interval-based forecasts, and clear justifications for each recommendation.

II. OBJECTIVES

The main objectives are:

- Develop a hybrid recommendation engine combining content-based filtering and collaborative filtering.
- Precompute analytics to ensure real-time query responses.
- Incorporate volatility, risk - adjusted returns (Sharpe ratio), CAGR, and forecasting with confidence intervals.
- Deliver explainable recommendations tailored to user profiles.
- Deploy an interactive dashboard for visualization and querying.

III. SCOPE AND USE CASES

A. Use Cases

- Retail investors: Personalized recommendations based on risk appetite.

- Wealth managers: Data - driven portfolio suggestions.
- Trading platforms: Transparent “recommended stocks” sections.

B. Example Scenarios

- Conservative investor: Large-cap, low-volatility stocks (e.g. - TCS, Infosys).
- Aggressive trader: High-volatility small - cap stocks (e.g.- Zomato, Paytm).
- Long-term investor: Forecast-driven growth recommendations.

IV. DATASETS

- **Stock Price Data:** Yahoo Finance (yfinance), Alpha Vantage (OHLVCV).
- **Company Fundamentals:** P/E ratio, Market Cap, Beta, Dividend Yield.
- **Simulated User Profiles:** Conservative / Balanced / Aggressive with sector preferences.

V. METHODOLOGY

A. Data Collection & Preprocessing

APIs (Yahoo Finance, Alpha Vantage) provide stock prices and fundamentals. Missing values are handled, features normalized, and structured storage created. Historical OHLVCV data is collected at daily resolution depending on module requirements.

B. Feature Engineering

Metrics such as daily returns, volatility, Sharpe ratio, and CAGR are computed and stored. Additional features extracted from fundamentals include market capitalization, ROE, ROA, profit margins and debt-to-equity.

C. Forecasting Future Prices

A hybrid approach using ARIMA, GARCH and LSTM is employed to obtain probabilistic forecasts (point predictions and confidence intervals).

D. Recommendation Engine

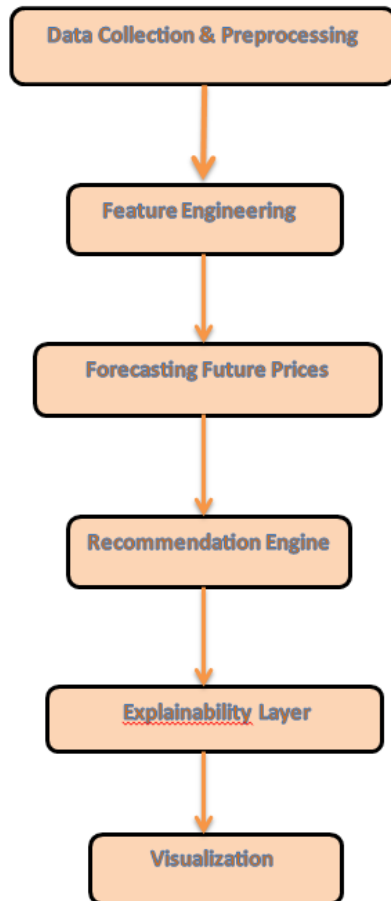
- Content-based: Match stocks to risk, sector, volatility, fundamentals.
- Collaborative: Matrix-factorization-like or item-item similarity based on simulated investor interactions.
- Hybrid: Combine content and collaborative candidates, then rank with forecast-aware risk-adjusted scoring.
- Ranking: Personalize final ranking according to user profiles (Conservative, Balanced, Aggressive).

E. Explainability Layer

Each recommendation includes compact justifications: volatility levels, Sharpe ratio, sector alignment, predicted mean/CI, and content-similarity evidence.

F. Visualization

Graphs for historical trends, volatility, and forecast confidence intervals are produced. Streamlit is used for rapid interactive deployment.



VI. DATA ENGINEERING AND STORAGE

This project uses real-time and historical market data fetched through Yahoo Finance (yfinance), covering 60+ frequently traded NSE stocks across major Indian market sectors such as Information Technology, Financial Services, Healthcare, Consumer Goods, Manufacturing, and Energy.

A. Data Cleaning

Price history (OHLCV) data often contains gaps due to non-trading days or API inconsistencies. Missing values were handled using a backward-fill strategy to maintain temporal continuity. Outliers introduced by corporate actions were smoothed after confirming corporate-action metadata (e.g., stock splits, dividends).

B. Feature Engineering

Multiple quantitative features were computed to characterize stock performance and risk:

- Daily and log returns
- Volatility (rolling standard deviation)
- Sharpe Ratio (return vs risk)
- CAGR (compounded annual growth rate)

Key fundamental features were extracted using the Yahoo Finance ticker metadata:

- Market capitalization
- Beta (systematic risk)
- Price-to-Earnings (P/E) ratio
- Dividend yield
- Profit margins
- Return on Equity (ROE) and Return on Assets (ROA)

C. Encoding and Normalization

Categorical attributes such as sector and industry were transformed using OneHotEncoding. All numeric attributes were rescaled to $[0, 1]$ using Min-Max normalization to eliminate scale bias in similarity modeling.

D. Output Storage

The engineering layer outputs:

- *metrics_data.csv*: risk-return performance metrics for ranking.
- *stocks_content_data.csv*: scaled feature representation for similarity learning.
- *models_data.pkl*: serialized model outputs (predictions, CIs, metrics) for fast serving.

VII. USER MODELING AND SYNTHETIC INVESTMENT DATA

Since real user investment portfolios are unavailable due to privacy restrictions, synthetic user data was generated to evaluate the system in a realistic manner.

A. Risk Profiles

Each user is assigned a risk tolerance level:

- Conservative (40%) – low volatility, stable returns
- Balanced (40%) – diversified with moderate volatility
- Aggressive (20%) – high volatility, high growth

B. Sector Preferences

Each simulated investor selects one or two preferred sectors from the available list, enabling realistic domain affinity and diversification.

C. Investment Probability Model

For a given stock i and user u , an investment likelihood score is computed as:

$$P(i|u) = 0.2 + f(\text{sector_match}, \text{risk}, \text{volatility}, \text{return})$$

where function f adds:

- Sector match bonus (+0.1).
- Risk-profile-driven adjustments based on volatility and CAGR:
 - For **conservative user** give more weights to low volatility stocks, so add:

$$0.25 \times (1 - \text{Volatility}) + 0.20 \times \text{mean-returns}$$

- For **balanced user** give equal weights to volatility and mean returns of stocks, so add:

$$0.20 \times (1 - |\text{Volatility} - 0.5|) + 0.20 \times \text{mean-returns}$$

- For **aggressive user** give more weights to higher mean return and higher CAGR stocks, so add:

$$0.30 \times \text{volatility} + 0.10 \times \text{mean-returns} + 0.10 \times \text{CAGR}$$

The final probability is clipped to $[0, 1]$ and top- k stocks are selected for each user ($3 \leq k \leq 10$). Resulting interaction matrix is:

$$U \in \{0, 1\}^{1000 \times 60}$$

The simulated dataset is exported as `users_investment_data.csv` for collaborative filtering experiments.

VIII. FORECASTING PIPELINE

Stock price forecasting plays a crucial role in predicting future market behavior and supporting risk-aware investment decisions. We design a hybrid forecasting pipeline integrating statistical and deep learning models to capture both linear and nonlinear temporal patterns.

A. Feature Preparation

To stabilize variance and convert non-stationary price series into stationary signals, log-returns are computed using multi-period differencing:

$$r_t = \log(P_t) - \log(P_{t-k}), \quad k \in \{5, 15, 30, 60\}$$

These features allow the models to learn both short-term and long-term temporal dependencies. A rolling window of 5 days is used for training, consistent with short-term trend dynamics in financial markets.

B. Model Training and Evaluation Framework

This section details the hybrid training pipeline used for stock forecasting and the evaluation methodology employed to select the most reliable model per stock. The pipeline integrates linear time-series models, deep learning models, volatility modeling, and a reliability-driven ensemble selection.

C. ARIMA Modeling

We fit an ARIMA(p, d, q) model to the stationary log-return series. A grid search is performed over:

$$p \in \{0, 1, 2, 3\}, \quad d \in \{0, 1, 2\}, \quad q \in \{0, 1, 2, 3\},$$

and the model with minimum Akaike Information Criterion (AIC) is selected:

$$(p^*, d^*, q^*) = \arg \min_{p, d, q} \text{AIC}(p, d, q).$$

The fitted ARIMA model provides in-sample fitted values and multi-step forecasts via the `get_forecast()` interface. In-sample residuals are computed for diagnostic testing.

D. LSTM-based Nonlinear Forecasting

We build an LSTM regression model using the Keras Functional API. Input features are multi-horizon differenced log-returns:

$$r_t^{(k)} = \log(P_t) - \log(P_{t-k}), \quad k \in \{5, 15, 30, 60\},$$

The architecture is tuned via grid search:

- Units: {64, 86, 128}
- Dropout: {0.1, 0.2}
- Epochs: 50–100
- Batch size: 16

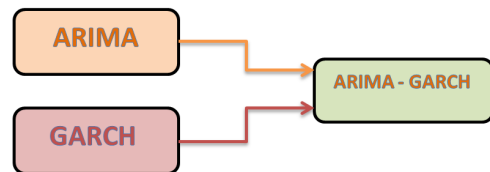
E. Residual Diagnostics and GARCH Enhancement

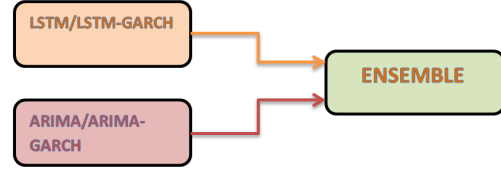
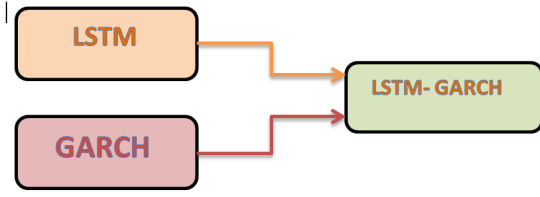
Residual checks include:

- Ljung–Box: autocorrelation
- Engle ARCH: heteroskedasticity

If ARCH effect exists ($p < 0.05$), GARCH(1, 1) is fitted:

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2.$$





F. Prediction to Returns and Confidence Intervals

Predicted log-returns are converted to returns:

$$R_t = \exp \left(\sum_{i=1}^t \hat{r}_i \right).$$

Confidence intervals leverage residual standard deviation or GARCH variance:

$$\text{Lower}_t = \exp \left(\sum_{i=1}^t \hat{r}_i - 1.96 \sqrt{\text{Var}_t} \right),$$

$$\text{Upper}_t = \exp \left(\sum_{i=1}^t \hat{r}_i + 1.96 \sqrt{\text{Var}_t} \right)$$

G. Evaluation Metrics

- RMSE
- MAPE
- Residual Standard Deviation
- Confidence Interval Coverage (CIC)

H. Reliability Score

A composite reliability score is computed to combine accuracy and calibration:

$$\text{Score} = w_1 \cdot \frac{1}{\text{RMSE} + \epsilon} + w_2 \cdot \frac{1}{\text{MAPE} + \epsilon} + w_3 \cdot \text{CIC} + w_4 \cdot \frac{1}{1 + \sigma_\epsilon}$$

Default weights used are $w_1 = 0.3$, $w_2 = 0.2$, $w_3 = 0.4$, $w_4 = 0.1$ and small ϵ for numerical stability.

I. Ensemble Forecasting Model

To jointly leverage:

- Linear pattern learning from ARIMA-GARCH
- Nonlinear learning from LSTM-GARCH

we create a reliability-weighted ensemble. Softmax normalization of model reliability scores determines fusion weights:

$$w_i = \frac{e^{\text{Score}_i}}{\sum_j e^{\text{Score}_j}}$$

Final ensemble predictions are computed as:

$$\hat{y}_{ens} = \sum_i w_i \hat{y}_i$$

Confidence intervals (upper/lower return bounds) are weighted similarly:

$$CI_{ens}(t) = \sum_i w_i \cdot CI_i(t)$$

J. Model Selection and Export

The model with highest Reliability Score per stock is saved for:

- Live prediction
- Recommendation ranking
- Streamlit visualization

Metadata is serialized to `datasets/models_data.pkl`.

IX. HYBRID EXPLAINABLE RECOMMENDATION ENGINE

The recommendation engine integrates three major components: content-based filtering, collaborative filtering, and risk-adjusted ranking based on forecasted performance.

A. Content-Based Filtering

Content-based filtering recommends stocks with similar financial, sectoral, and risk characteristics to those already in the user portfolio. Each stock is represented as a high-dimensional feature vector $\mathbf{x}_i$ integrating fundamentals, risk factors, and categorical encodings. Cosine similarity is used as the similarity metric:

$$\text{Sim}(i, j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|}$$

Nearby neighbors in this space represent companies with comparable risk-return profiles and fundamentals.

1) *Feature Space Design*: To generate meaningful stock similarity and recommendations, a comprehensive set of features is used to construct the similarity vector. These features capture multiple dimensions of a firm's characteristics, including its financial health, risk exposure, performance metrics, sectoral and industry classification, and geographic presence. Each feature group is described below in detail.

a) *Valuation & Financial Health Metrics*: This group captures a company's financial strength, profitability, and capital structure. These metrics help in evaluating fundamental value and sustainability:

- **marketCap**: Market capitalization represents the total value of a company's outstanding shares. It reflects firm size, investment scale, and market perception. Larger firms generally offer stability, while smaller firms may provide higher growth potential.
- **returnOnEquity (ROE)**: ROE measures the profitability relative to shareholders' equity. It indicates how efficiently a company is using equity capital to generate profits.

- **returnOnAssets (ROA)**: ROA evaluates the efficiency of asset utilization in generating profits.
- **profitMargins, grossMargins, ebitdaMargins**: Operational profitability metrics at different stages of the cost structure.
- **debtToEquity**: Measures financial leverage and solvency risk.

b) *Risk Factors*: Risk-related metrics provide insight into the company's volatility and governance quality:

- **beta**: Sensitivity to overall market movements.
- **auditRisk, boardRisk, compensationRisk, shareholderRightsRisk, overallRisk**: Corporate governance indicators assessing operational and managerial risks.

c) *Performance and Return Statistics*:

- **CAGR**: Long-term compounded annual growth rate.
- **Volatility**: Measures market and downside risk exposure.
- **Mean_Return**: Average historical return.
- **Sharpe_Ratio**: Risk-adjusted return efficiency.

d) *Sector (One-Hot Encoding)*: Sectors classify companies based on broad economic activity:

- Basic Materials, Technology, Financial Services, Utilities, Consumer Cyclical, Consumer Defensive, Energy, Healthcare, Industrials, Communication Services.

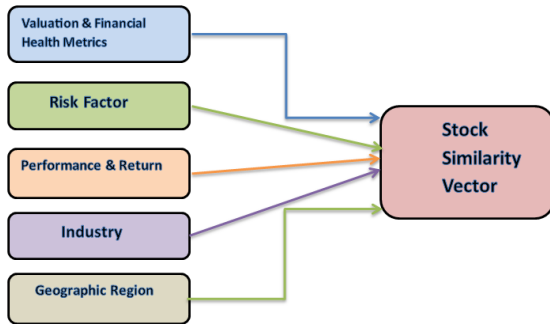
e) *Industry (One-Hot Encoding)*: Industry classification provides finer granularity:

- Examples: Software Infrastructure, Drug Manufacturers, Auto Manufacturers, Steel, Oil & Gas, Packaged Foods, etc.

f) *Geographic Region (Encoding)*:

- Region indicators: **region_US, region_EU, region_APAC, region_None**.

These multi-criteria features collectively ensure that the stock similarity vectors reflect a balance between fundamental strength, governance quality, historical performance, sectoral alignment, industry peers, and geographic exposure.



B. Collaborative Filtering

A user-stock binary interaction matrix supports item-item similarity-based collaborative filtering:

$$Score(i|u) = \sum_j Sim(i, j) \cdot I(u, j)$$

where $I(u, j)$ is the binary indicator of user u investing in stock j . This identifies stocks preferred by similar investor profiles.

C. Risk-Adjusted Forecast Ranking

To ensure personalized investment suitability, each stock is ranked using a scoring function integrating forecasts and volatility. The score blends predicted mean returns and CI-bounds with volatility and user risk profile.

1) *Scoring Logic*: A base score is computed:

$$Pred_Score_i = 0.5 \cdot \hat{R}_i + 0.25 \cdot \hat{L}_i + 0.25 \cdot \hat{U}_i$$

with adaptive adjustments:

- **Conservative**: prioritize downside protection

$$Pred_Score_i = \left(0.5\hat{R}_i + 0.35\hat{L}_i + 0.15\hat{U}_i \right) \cdot (1 - V_i)$$

- **Balanced**: prefer moderate volatility

$$Pred_Score_i = Pred_Score_i \cdot (1 - |V_i - 0.5|)$$

- **Aggressive**: reward growth potential and volatility

$$Pred_Score_i = \left(0.5\hat{R}_i + 0.15\hat{L}_i + 0.35\hat{U}_i \right) \cdot V_i$$

Stocks are ranked by $Pred_Score$ and the top- k are returned to the user.

D. Cold-Start Handling

For new users with no portfolio history, recommendations rely on risk-aware ranking across all stocks. For new stocks with limited interaction data, content-based similarity ensures appropriate targeting.

X. INTERACTIVE WEB APPLICATION USING STREAMLIT

To enable practical user interaction with the proposed recommendation engine, we developed a fully functional web application using Streamlit. The application offers personalized recommendations and visualization-driven explainability in a user-friendly interface.

A. Page Layout and Design

The webpage is initialized with customized configuration:

- **Title**: AI-Powered Stock Recommendation System
- **Theme**: Wide layout with dark-blue accent visuals
- **Styling**: Custom CSS for gradients, card UI, modern typography, hover animations

Figure 1 shows the home screen section including the title banner and introductory guidance.

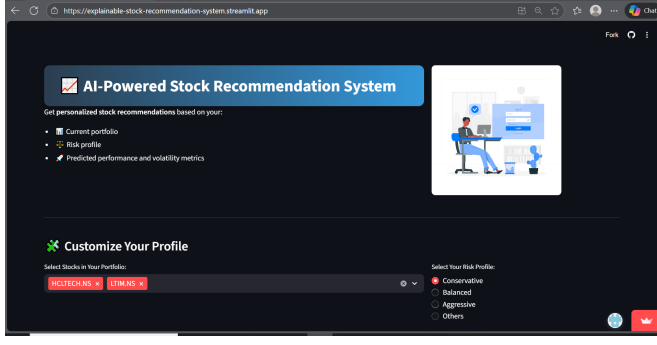


Fig. 1: Home page layout of Streamlit Web Application

B. User Input Configuration

Users specify:

- Current stock portfolio via multi-select dropdown
- Risk appetite: Conservative / Balanced / Aggressive / Others

These inputs feed into hybrid filtering and ranking modules.

C. Personalized Stock Recommendations

Upon clicking the *Get My Recommendations* button:

- If no portfolio is provided → purely forecasting + risk-based ranking
- If portfolio is provided → hybrid similarity + forecast-driven ranking

Top-5 picks are shown with key metrics (Mean, Lower, Upper returns, Volatility).

Symbol	mean_returns	lower_returns	upper_returns	Volatility
PAYTM.NS	1.128722	1.087636	1.171952	0.022948
SIN.NS	1.076493	1.060247	1.083739	0.044860
INFY.NS	1.068857	1.059519	1.077796	0.023020
TCS.NS	0.951498	0.972855	1.030531	0.022270
IDECC.NS	0.904561	0.907564	1.003007	0.031110

Fig. 2: Recommended Stocks Table with Forecast Metrics

D. Interactive Explainability Dashboard

For each recommended stock, four visualization tabs are provided.

1) *Performance Chart*: To provide insights into the past market performance of a stock, the historical daily closing prices are visualized using a time series plot. This helps investors understand market stability, volatility patterns, and momentum trends.

Let P_t denote the closing price at time t , where:

$$t = 1, 2, \dots, N$$

The plotted dataset can be expressed as:

$$\text{Trend} = \{(t, P_t)\} \quad \forall t \in [1, N]$$

This visualization enables detection of:

- Short-term and long-term market trends
- Seasonal or cyclical patterns
- Sudden price jumps due to external events
- High vs. low volatility periods

It assists in preliminary technical evaluation before observing model forecasts.

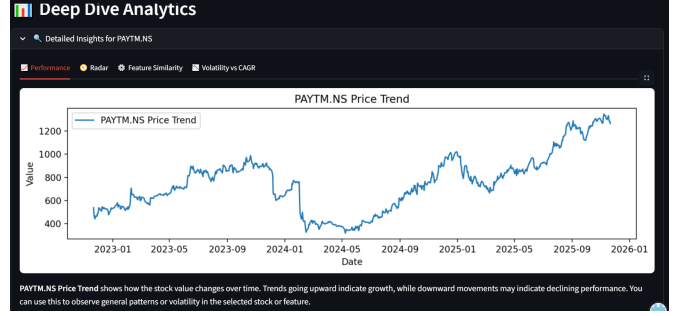


Fig. 3: Historical Price Trend View

2) *Radar Chart*: A radar chart offers a compact comparison of stock quality across multiple key performance indicators (KPIs). We consider a vector of normalized investment factors:

$$\mathbf{v} = [\text{CAGR}, \text{Volatility}, \text{Sharpe Ratio}, \text{Mean Return}]$$

Each value is scaled using min-max normalization for fair representation:

$$v'_i = \frac{v_i - \min(v_i)}{\max(v_i) - \min(v_i)}$$

The radar plot summarizes:

- Aggressiveness vs. defensive nature of the stock
- Risk-return balance efficiency (Sharpe Score)
- Comparative positioning against alternative stocks

A larger enclosed area indicates stronger overall investment characteristics.

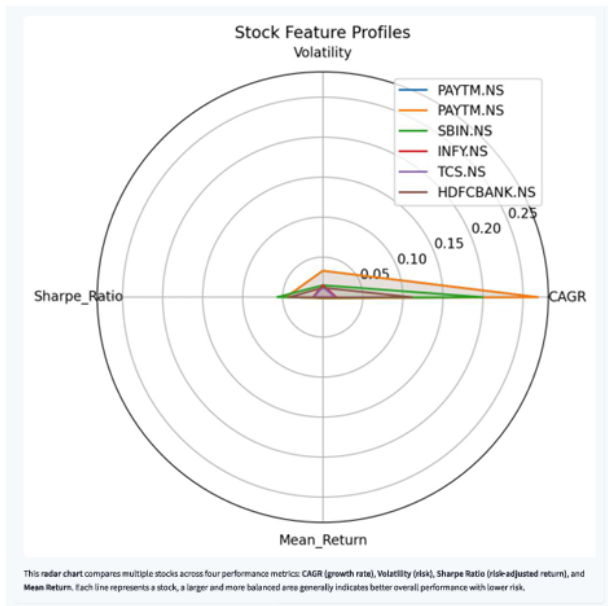


Fig. 4: Radar Chart for Risk-Return Attributes

3) *Feature Similarity*: To enhance transparency in recommendations, a similarity heatmap highlights stocks with comparable fundamentals and performance metrics.

Each stock i and j is represented by a feature vector x including:

- Risk indicators: Volatility, Beta
- Profitability metrics: ROE, ROA, Profit Margins
- Sector and Industry encodings (One-Hot)
- Return-based factors: CAGR, Mean Return, Sharpe Ratio

The top- k similar peers are selected as:

$$\text{Similar Stocks} = \text{TopK}(\text{Sim}(i, \text{others}))$$

This helps investors discover diversification-friendly but behaviorally similar stocks.

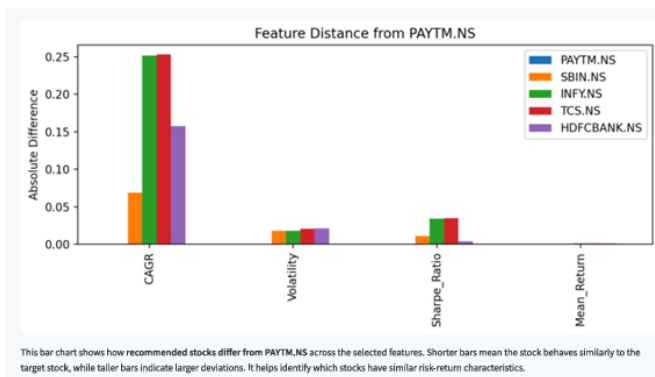


Fig. 5: Feature Similarity Map for Explainability

4) *Volatility vs CAGR Visualization*: Investment suitability is assessed visually through a scatter plot of:

- X-axis: Volatility (risk)
- Y-axis: CAGR (return)

According to risk-reward theory:

Ideal Picks: High CAGR $\wedge$ Low Volatility

The recommended stock is highlighted as:

$$i^* = \arg \max_i \text{Pred_Score}(i)$$

Investment interpretation based on quadrant location:

- High Return - Low Risk: Best fit for most investors
- High Return - High Risk: Good for aggressive profiles
- Low Return - Low Risk: Suited for conservative investors
- Low Return - High Risk: Generally avoidable

This visualization provides clear understanding of reward potential vs. exposure to risk, tailored to investor preference.

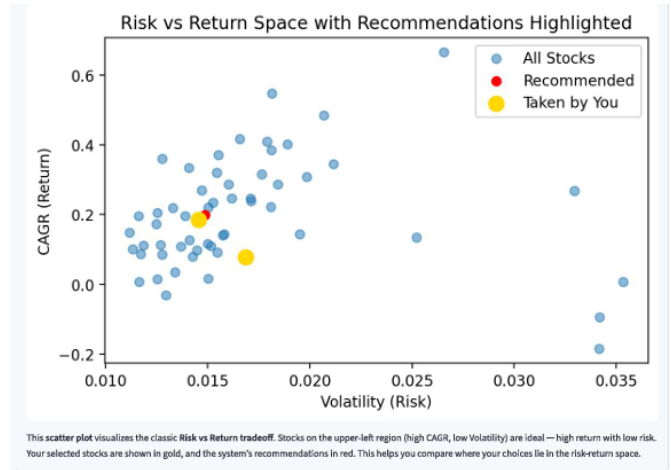


Fig. 6: Volatility vs CAGR Trade-off Plot

E. Deployment Outcome

The Streamlit interface ensures:

- Real-time model inference
- Clearly interpretable reasoning behind each recommendation
- Beginner-friendly investment decision assistance

F. Explainability Layer

For each recommendation, the system displays:

- Historical price performance visualization
- Forecasted price range with confidence intervals
- Risk metrics such as volatility and Sharpe ratio
- Textual justification aligned with user profile

XI. TOOLS AND TECHNOLOGIES

- **Data Collection**: yfinance, Alpha Vantage
- **ML Models**: Scikit-learn, LightFM
- **Forecasting**: ARIMA, GARCH, LSTM
- **Visualization**: Matplotlib, Seaborn, Plotly
- **Backend**: Python, Flask/FastAPI
- **Deployment**: Streamlit

XII. OUTCOMES

The proposed Explainable Stock Recommendation System successfully integrates hybrid forecasting models, risk-aware personalization, and real-time interactive visualization to support data-driven retail investment decisions. The following key outcomes were achieved:

A. Accurate and Reliable Forecasting

The hybrid ARIMA-LSTM-GARCH ensemble demonstrated improved prediction stability by capturing both linear autoregressive patterns and nonlinear market movements. The Reliability Score framework ensured that the best model was automatically selected for each stock based on:

- Predictive accuracy (RMSE, MAPE)
- Volatility calibration (Residual Standard Deviation)
- Confidence Coverage (percentage of actuals within interval bounds)

B. Personalized and Explainable Recommendations

The system generated tailored stock rankings based on the user's:

- Existing portfolio allocation
- Risk appetite (Conservative, Balanced, Aggressive)
- Sector affinity and return expectations

Each recommendation is accompanied with visualization and textual explanations.

C. Efficient Real-Time Analytics

To enable seamless deployment:

- All metrics such as volatility, Sharpe ratio, forecast returns, and similarity scores were precomputed and serialized.
- The Streamlit-based UI efficiently serves recommendations with negligible latency.

D. Cold-Start Problem Mitigation

The system handles both new users (risk-profile-driven recommendations) and new stocks (content-based recommendations), ensuring robustness under data sparsity.

E. Visual Explainability for Better Decision Support

The interface includes multiple analytical visualizations that equip investors with clarity in decision-making:

- Price trends for temporal understanding
- Feature similarity maps for diversification
- Risk-return scatter for portfolio balancing

F. Potential for Real-World Fintech Adoption

The project demonstrates practical deployability in:

- Retail trading apps
- Robo-advisory investment platforms
- Wealth management analytics dashboards

Scalability is achievable by extending the training pipeline to more stocks and incorporating real user behavior data.

XIII. CONCLUSION

This project successfully demonstrates an explainable AI-based stock recommendation system that integrates:

- Hybrid forecasting using ARIMA-LSTM-GARCH ensemble models
- Personalized recommendations based on user risk profiles
- Interactive visual analytics for explainability
- Real-time system response through precomputed metrics

The system effectively bridges the gap between advanced predictive modeling and transparent investment guidance. By incorporating volatility-aware forecasting and confidence-interval-based ranking, the proposed solution enhances user trust and informed decision-making in retail finance.

XIV. FUTURE WORK

While the current version performs effectively, several enhancements are planned to elevate practical applicability:

- Integration of real-world user portfolios using secure brokerage APIs
- Inclusion of macroeconomic indicators such as inflation, interest rates, and market indices
- NLP-based analysis of financial news and sentiment from earnings reports
- Deployment with cloud-based auto-refresh of analytics and forecasts

XV. LIMITATIONS

The current system, while robust, has a few constraints:

- Forecasts are sensitive to high-volatility market conditions and black-swan events
- Confidence intervals rely on historical residual variance, assuming distribution stability
- Absence of real investor behavioral data restricts collaborative filtering accuracy
- Recommendations do not fully consider liquidity or transaction costs

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